No Kink at R1: The CVE-Publication Surge Localizes to Late 2025, Not the DeepSeek-R1 Release

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analysis pass of record: 2026-07, `natex` v0.2.0, seed 0

Abstract

Monthly CVE publications did not bend at DeepSeek-R1. On 90 months of National Vulnerability Database publication counts (January 2019 – June 2026), a sharp regression kink design in calendar time at the R1 release (20 January 2025) estimates a slope change in ln monthly publications of +0.038 ln-points/yr (HC1 se 0.128, $z = 0.30$, $p = 0.764$, 95% CI $[-0.213, +0.289]$; 18 pre / 17 post months, bandwidth 1.5 years, triangular kernel). A scout-flagged +0.18 ln/yr break — identical to the uniform-kernel cell (+0.184, se 0.104, $p = 0.077$; Newey–West $p = 0.129$ at 6 lags) — is correctly identified as an artifact of action far from the cutoff: the estimate grows with bandwidth and vanishes under triangular weighting; a 49-cutoff placebo grid shows HC1 is oversized on this smooth serially correlated series (empirical size 0.14 at nominal 0.05) and puts the placebo-calibrated p at 0.80 (0.76 on the 37-cutoff clean subgrid), with 38 of 49 placebos exceeding the observed |kink| and the largest (+0.385, $z = 3.5$) at November 2020, nowhere near any AI release; a local quadratic flips the contrast negative (-0.90, $p = 0.015$ at bandwidth 2.0, uniform); and sub-window slopes place the first nine post-R1 months at +0.103 ln/yr — *shallower* than the +0.280 pre-trend — with all steepening from November–December 2025 onward (+1.19 ln/yr through June 2026). Blind LoRD3 localization (seed 0, $k \in \{12, 24, 36\}$, discovery reads only the series and time) independently ranks November/December 2025 (scan $p = 0.02/0.01$ at $k = 24/36$) and February–April 2026 first; R1's month appears in no top-20 at any k. An o1-preview contrast is a clean null (-0.035, $p = 0.844$). The post-2025 acceleration is real — 3,116 publications in November 2025, 5,643 in December, 8,001 by June 2026 — but it arrives 10–17 months after the release, consistent with NVD operational disruption and backlog catch-up on top of CNA-program expansion. On this series, attribution of the surge to AI-assisted vulnerability discovery is unsupported; the result stands as a calibrated null plus an artifact-identification case study, and reserve-date robustness via the `cvelistV5` history is required before any attribution claim.

1 Introduction

DeepSeek released R1, the first open-weights frontier reasoning model, on 20 January 2025 [4]. The release is a natural candidate for a regime change in the economics of vulnerability discovery: contemporaneous work showed that LLM agents can autonomously exploit real one-day vulnerabilities from their CVE descriptions [5], and an open-weights reasoning model removes the API chokepoint that had kept such capabilities meterable. If AI assistance materially accelerated vulnerability

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discovery *at scale*, the aggregate flow of published CVEs should steepen — kink — at or shortly after the release. This paper tests that hypothesis directly.

The series demands discipline before any such reading. CVE publication counts are heavily process-driven: annual volumes are forecastable to within roughly 8% a year ahead from disclosure-pipeline variables alone [9]; the publication dates and other fields of the National Vulnerability Database have documented quality problems [1]; and NVD processing itself broke and recovered inside the sample window — the February 2024 enrichment slowdown and subsequent backlog, followed by operational changes that NIST announced in April 2026 in response to record CVE growth [11]. Publication-date slopes therefore conflate discovery with processing, and any slope break must be localized before it is attributed.

The estimand — a change in trend slope at a known date — is the regression kink design of [3], applied with calendar time as the running variable; [2] formalize the closely related difference-in-kinks design that the same toolkit implements. Two methodological strands discipline the exercise. Regression discontinuity and kink designs in *time* lack the randomization logic of cross-sectional designs: serial dependence and smooth trends generate spurious breaks [7]. And kink estimates on smooth series are known to produce spuriously significant slope changes, which is why placebo distributions at non-event dates are the appropriate yardstick [6]. Both concerns are operationalized below — a 49-cutoff placebo grid whose calibrated p-value is the inference of record, curvature checks, HAC variants, and a blind localization scan — estimated with the `natex` toolkit [8].

The verdict is a clean, well-powered null at the cutoff, plus a correctly identified artifact: the post-2025 acceleration in CVE publications is real, but every localization tool places it in late 2025 and 2026, ten to seventeen months after R1, with the first nine post-release months *shallower* than the pre-trend.

2 Data

The outcome series is monthly counts of CVEs by NVD publication date, obtained from the NVD API 2.0 [10] as `totalResults` per month-long `pubStartDate/pubEndDate` window (fetched July 2026): 90 months, January 2019 through June 2026, with no missing months. The running variable is mid-month decimal time $t = \text{year} + (\text{month} - 0.5)/12$, and the outcome is ln of the monthly count. The R1 cutoff is $t = 2025.055$ (20 January 2025); the o1-preview contrast cutoff is $t = 2024.70$ (12 September 2024). Levels anchor the story told below: November 2025 has 3,116 publications — the series' anomalous dip — followed by 5,643 in December 2025 and 8,001 by June 2026. The series records the NVD *published* date, not the CVE reserve date; the distinction carries the paper's principal caveat (Section 5).

3 Design and Methods

Sharp RKD in calendar time. The estimand is the change in the slope of ln monthly publications, in ln-points per year, at the cutoff, estimated by local linear regression on each side within the bandwidth (`natex` v0.2.0 [8], `natex kink` with a unit policy-kink change; [3, 2]). The headline specification uses bandwidth 1.5 years, triangular kernel, degree 1, donut 0 ($n = 35$; 18 left, 17 right). A 48-cell grid varies cutoff $\in$ {R1, o1} $\times$ bandwidth $\in \{1.0, 1.5, 2.0\}$ years $\times$ kernel $\in$ {triangular, uniform} $\times$ donut $\in \{0, 1\}$ months $\times$ degree $\in \{1, 2\}$. The uniform bandwidth-1.5 cell reproduces the wave-2 scout's confirmed side-OLS slope delta ($+0.280 \to +0.464$ ln/yr) exactly, and a fully interacted local WLS mirror of `natex` matches its uniform kink to 10^{-8}. The design is deterministic; seed 0 is declared and no stochastic step exists in estimation.

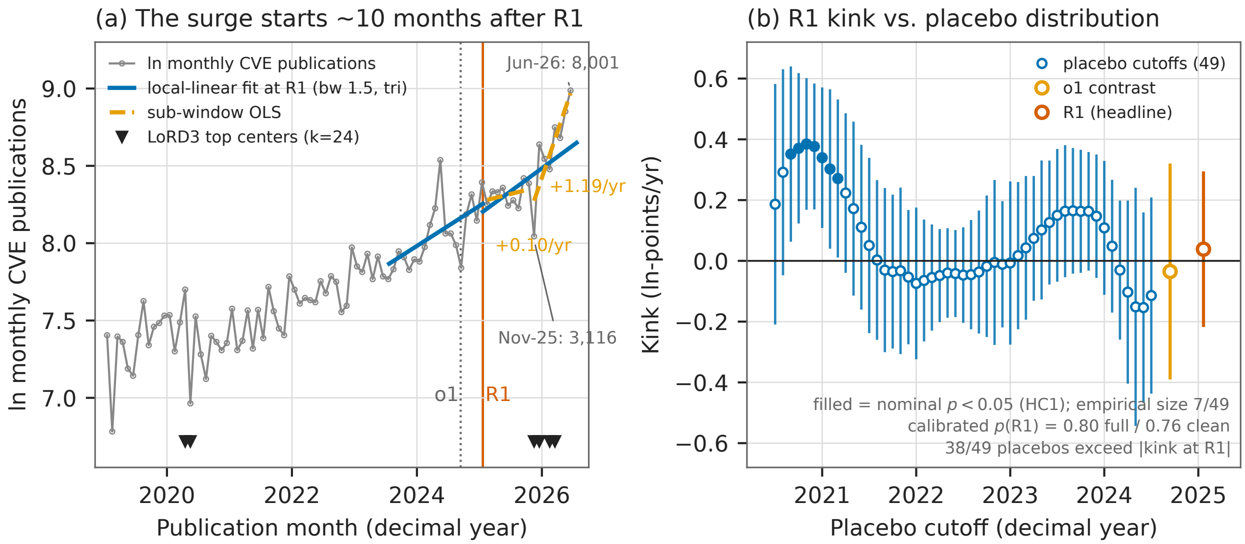

Figure 1: (a) ln monthly CVE publications (gray), 2019-01 – 2026-06, with per-side local-linear fits at the R1 cutoff (blue, bandwidth 1.5 years, triangular kernel): the slope change is +0.038 ln/yr (p = 0.764). Orange dashed segments are sub-window OLS fits: the first nine post-R1 months run at +0.103 ln/yr, *shallower* than the +0.280 pre-trend; November 2025 – June 2026 runs at +1.19 ln/yr. Markers along the bottom are the top blind LoRD3 discovery centers ($k = 24$): November/December 2025 and February–April 2026, plus the April/May 2020 COVID dip — never R1. The November 2025 dip (3,116) and December spike (5,643) are labeled. (b) The same specification re-estimated at 49 monthly placebo cutoffs (2020.5–2024.5), with 95% HC1 intervals; filled markers are nominally significant at the 5% level (7 of 49 — empirical size 0.14). The R1 estimate (vermillion) sits well inside the placebo distribution: 38 of 49 placebos exceed it in magnitude and the placebo-calibrated p is 0.80; the largest placebo (+0.385, z = 3.5, November 2020) is nowhere near any AI release. The o1 contrast cell (orange) is a clean null.

Inference on a smooth serially correlated aggregate. HC1 treats months as independent; on this series it is demonstrably oversized (residual lag-1 autocorrelation 0.23 in the uniform cell), so three guards are run. *(i) HAC:* Newey–West standard errors (Bartlett, lags 3/6/12) on the fully interacted local fit, with and without month-of-year dummies. *(ii) Placebo calibration*, in the spirit of [6]: the headline specification re-estimated at 49 monthly placebo cutoffs (2020.5–2024.5), reporting the empirical size at nominal 0.05 and the placebo-calibrated $p = (1 + \#\{|z_{\text{placebo}}| \geq |z_{\text{R1}}|\})/(1 + 49)$; a clean 37-cutoff subgrid keeps only placebo windows that end before R1. The calibrated p is the inference of record. *(iii) Curvature:* degree-2 refits test whether any kink survives allowing a smooth quadratic [7]. *Blind localization:* `natex discover` (LoRD3, seed 0, $k \in \{12, 24, 36\}$) scans the series with discovery reading only $(t, \ln n)$ — never the cutoff — and ranks candidate break locations; if the R1 date does not surface, the case for an R1-localized break must come from priors, not the data.

4 Results

Null at the cutoff, with power. The headline kink at R1 (Table 1, Panel A) is +0.038 ln-points/yr (HC1 se 0.128, z = 0.30, p = 0.764, 95% CI $[-0.213, +0.289]$): the local trend in ln publications does not change measurably at the release. The o1-preview contrast cell — same specification at $t = 2024.70$ — is likewise null (-0.035, se 0.179, $z = -0.20$, $p = 0.844$), matching the scout's confirmed side-OLS delta of -0.028. The null is not for lack of power: the same specification moved across 49 placebo cutoffs rejects nominally at 7 of 49 and detects a $+0.385$ break ($z = 3.5$) at November 2020, so the instrument registers breaks of the size at issue when they exist.

The uniform $+0.18$ is an edge artifact, and triage identifies it as one. The uniform-kernel cell — numerically identical to the scout's confirmed side-OLS delta ($+0.280 \to +0.464$) — gives $+0.184$ (se 0.104, $p = 0.077$), and is not significant under HAC (Newey–West $p = 0.129$ at 6 lags, 0.077 at 12). Four facts disqualify an R1-localized reading. First, the kernel/bandwidth pattern: the estimate vanishes under triangular weighting (which concentrates mass near the cutoff) and *grows* with bandwidth under uniform weighting ($+0.068$ at bandwidth 1.0, $+0.184$ at 1.5, $+0.217$ at 2.0) — the signature of action far from the cutoff, not at it. Second, placebo calibration: HC1's empirical size on this series is 0.14 at nominal 0.05 (7/49; the oversizing catalogued as `natex` issue #51), 38 of 49 placebos exceed the observed |kink|, and the calibrated p is 0.80 on the full grid and 0.76 on the clean 37-cutoff subgrid. Third, curvature: degree-2 refits flip the R1 contrast *negative* (-0.77 at bandwidth 1.5 triangular; -0.90, $p = 0.015$, at bandwidth 2.0 uniform) — a local quadratic absorbs the "kink" entirely. Fourth, the nominally significant stragglers (uniform-with-donut cells at $p = 0.011 - 0.033$; seasonal Newey–West uniform cells at $p = 0.001 - 0.033$, the latter spending 15 parameters on 35 observations) all inherit the same edge-driven uniform estimate and are overridden by the calibration and the localization evidence.

Where the action actually is. Sub-window OLS slopes decompose the post-period: the pre-R1 window runs at $+0.280$ ln/yr; the first nine post-R1 months (February–October 2025) run at $+0.103$ — *shallower* than the pre-trend; November 2025 – June 2026 runs at $+1.19$ ln/yr. All steepening starts roughly ten months after the release. Year-over-year log growth is ≈ 0.30 both in the twelve months before R1 and in the last observed twelve months — the surge restores the trend as much as it breaks it. Blind LoRD3 localization agrees: at $k = 24$ the top discovery centers are November/December 2025 (scan $p = 0.02$), then February–April 2026; at $k = 36$ the same centers lead (scan $p = 0.01$); at $k = 12$ the late-2025/2026 cluster leads with the April/May 2020 COVID-era dip next; R1's month appears in no top-20 at any k. The dip-then-surge shape — November 2025 at 3,116 publications (fit residual -0.45, the series' anomaly), December at 5,643, June 2026 at 8,001 — is consistent with an NVD operational disruption followed by backlog catch-up (the October–November 2025 federal shutdown is the obvious candidate) on top of continuing CNA-program expansion [11]; that attribution is a hypothesis consistent with the blind localization, not a result verified in this run.

5 Caveats and Conclusion

Five caveats bound the claims. First, and principally: this is an NVD *published-date* series. Backlog clearance mechanically shifts counts late — the February 2024 NVD crisis and the 2025–26 catch-up are both in-sample — so publication-date slopes conflate discovery and processing; reserve-date robustness via the `cvelistV5` git history (`dateReserved` by CNA, with the Linux-kernel CNA split out) is mandatory before any attribution claim, and this run did not perform it. Second, the attribution of the November 2025 dip and December 2025 spike to the October–November 2025 federal shutdown plus NVD catch-up is a hypothesis consistent with the blind-scan localization, not verified in-run. Third, HC1 — and every nominal p-value here — is oversized on this smooth serially

Table 1: Sharp RKD in calendar time on ln monthly CVE publications. Headline specification: cutoff R1 ($t = 2025.055$), bandwidth 1.5 years, triangular kernel, degree 1, donut 0, HC1.

Cell	Kernel/spec	Kink	se	z	p
Panel A: headline cells (ln-points/yr; bandwidth 1.5, degree 1)					
R1 ($t = 2025.055$), headline	triangular	+0.038	0.128	0.30	0.764
R1, scout's cell	uniform	+0.184	0.104	1.77	0.077
R1, Newey–West (6 / 12 lags)	uniform	+0.184	—	—	0.129 / 0.077
o1 ($t = 2024.70$), contrast	triangular	−0.035	0.179	−0.20	0.844
Panel B: triage of the uniform estimate					
R1, uniform, bandwidth 1.0 / 2.0	uniform	+0.068 / +0.217	—	—	0.769 / 0.022
R1, degree 2, bw 1.5	triangular	−0.773	0.529	−1.46	0.144
R1, degree 2, bw 2.0	uniform	−0.900	0.369	−2.44	0.015
Largest placebo (Nov 2020)	triangular	+0.385	0.109	3.54	0.0004
Panel C: placebo calibration of the R1 cell (headline spec)					
Empirical size at nominal 0.05	49 cutoffs	0.14 (7/49; `natex` issue #51)			
Placebo-calibrated p, full / clean	$\lvert z \rvert$ rank	0.80 (49 cutoffs) / 0.76 (37 clean)			
Placebos exceeding observed $\lvert$kink$\rvert$	—	38 of 49			

Notes: kinks in ln-points per year; HC1 standard errors against standard normal critical values unless noted. The placebo grid re-estimates the headline specification at 49 monthly cutoffs (2020.5–2024.5); the clean subgrid keeps the 37 cutoffs whose windows end before R1. Sub-window OLS slopes (ln/yr): pre-R1 +0.280; February–October 2025 +0.103; November 2025 – June 2026 +1.19. Blind LoRD3 top centers: November/December 2025 (scan $p = 0.02$ at $k = 24$, 0.01 at $k = 36$), then February–April 2026 and April/May 2020; R1 in no top-20 at any k.

correlated aggregate (empirical placebo size 0.14 at nominal 0.05; `natex` issue #51): the placebo-calibrated inference is the inference of record, and the few uniform-with-donut and seasonal-HAC cells at nominal $p = 0.001$–0.033 inherit the same edge-driven uniform estimate and are overridden by calibration and localization. Fourth, calendar time is the running variable: any event co-timed with January 2025 is confounded by construction [7] — moot given the null at the cutoff, but binding for any future positive claim — and this is a single national series with no cross-sectional contrast. Fifth, the post window is truncated at June 2026 (17 post- versus 18 pre-months), and late placebo-cutoff windows unavoidably overlap the post-R1 region, which is why the clean 37-cutoff subgrid is reported alongside the full grid.

The conclusion is a calibrated null plus an identified artifact. At the DeepSeek-R1 release the ln CVE-publication trend shows no kink (+0.038, $p = 0.764$; calibrated $p = 0.80$) on a design with demonstrated power; the o1 contrast is equally null. The scout-flagged +0.18 is real arithmetic but wrong attribution: it is the uniform kernel handing the window edges full weight, and every localization tool — kernel/bandwidth pattern, sub-window slopes, curvature absorption, placebo calibration, and a seed-0 blind scan that never sees the cutoff — places the action at November/December 2025 through 2026, ten to seventeen months after the release, with the first nine post-R1 months *shallower* than the pre-trend. Attribution of the surge to AI-assisted vulnerability discovery [5, 4] is unsupported on this series. The result is promotable only as an artifact-identification and backlog case study — the dip-then-surge is consistent with NVD operational disruption and catch-up on top of CNA expansion [11] — and, per the collection's rules, any attribution-bearing follow-up must start from reserve dates, not publication dates.

Reproducibility. All estimates were produced with `natex` v0.2.0 [8]; estimation is deterministic (local weighted least squares), seed 0 was declared, and the only stochastic step anywhere in the pipeline is LoRD3's seeded scan calibration. The NVD monthly counts are fetched by a committed script in the run of record and are not committed to the repository. Figure 1 regenerates deterministically from the committed `figures/make_fig.py`, which asserts the headline estimates, the recomputed sub-window slopes, the placebo-calibration quantities, and the LoRD3 center ranks against the numbers of record before drawing.

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